

Some Competing Products Available in the Market

Features	Yuktix UDAC	FlyportPRO module and IoT kit	GrovePi	Re-Mote	Wasp mote
Sensors	- Multiple - Plug-and-play	- Multiple - Plug-and-play	- Multiple - Plug-and-play	- Built-in temperature sensor - Multiple external sensor integration	- Built-in temperature sensor and accelerometer - Multiple external sensor integration
Communication	Ethernet, RS485, GSM and/or ZigBee	Wi-Fi, GPRS or Ethernet	To Raspberry Pi	2.4GHz IEEE 802.15.4 radio with 868/915MHz RF transceiver	Ethernet, Wi-Fi and/or 3G/ GPRS
Memory	Onboard SD card	Internal and external flash memory	No	Onboard micro-SD card	No
Power	Solar panel, battery and/or mains	5V, 0.5/2A	No	Built-in battery charger to connect to solar panel or rechargeable batteries	Battery
Cloud storage	Yes	No	No	No	Yes
Support	Dedicated support team	Community support	Community support	Dedicated support team	Support forum
Availability	For sale	For sale	For sale	Pre-order	For sale
Price	Application-specific	US\$ 64 to US\$ 75	US\$ 29.99 to US\$ 89.99	US\$ 70	Dependent

much help to an inexperienced user. Cost of programming often surpasses the cost of the development platform itself. Moreover, the quality guarantee of some software is very weak compared to the hardware and can result in escalating maintenance. This is when the team decided to provide customers with a stable and simple firmware along with Yuktix board.

Designing such a device that covers multiple domains was a real challenge, according to the team. Understanding and covering the sensor and protocol requirements, providing both off-grid and grid power solutions, developing the hardware and a supporting PCB, and placing the radio module on the board without introducing noise to the measurements, were to be achieved keeping the form factor into consideration. Developing such an interoperable device from scratch was tough, and it took about four iterations and 1.5 years to get the device to the market.

Why Yuktix

There are many IoT boards available in the market. These only address a part of the solution provided by Yuktix. With such boards, cost of programming and risk cannot be ascertained beforehand. Also, IoT

Would Yuktix suit my application

Yuktix UDAC is not specific to an application. The company provides domain-specific customised solutions around its core intellectual property, which includes hardware, firmware stack and cloud stack. A network of automatic weather stations powered by Yuktix has already been installed across Bengaluru.

Here are a few other examples of how Yuktix can be deployed:

1. Industrial equipment that currently communicate over Modbus can be monitored and controlled remotely
2. GSM capability can be introduced to existing RFID readers
3. LED streetlights can be controlled by deploying local wireless networks

boards primarily target developers and DIY makers. Typical DIY makers and hobbyists never consider the time and efforts. However, the same is crucial for corporations' budgets.

According to its developers, this product is the only end-to-end solution available for industry-level IoT projects. With out-of-the-box solutions for firmware, data capture, communication protocols, cloud connectivity, server layer and Android application provided along with UDAC platform, these offer a comprehensive solution to its customers.

The high-end remote terminal units (RTUs) will do about 20 per cent and wireless data loggers will do 10 per cent of what Yuktix offers. But the issue here is that these are not plug-and-play solutions. With similar solar-powered devices, the user would be constrained with a specific

About the product

Product : Yuktix UDAC
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set of sensors. Also, the reprogramming and solution development cost for all such offerings would be very high, says the team. But Yuktix provides the firmware and mobile application, thereby saving the cost of programming.

Future releases

Next version of Yuktix will have over-the-air (OTA) features and would implement chips having security as part of the core. The team also plans to remove the power supply module, stack the interfaces and make it plug-and-play to reduce the size. ●